

Option D – Energy and the Environment

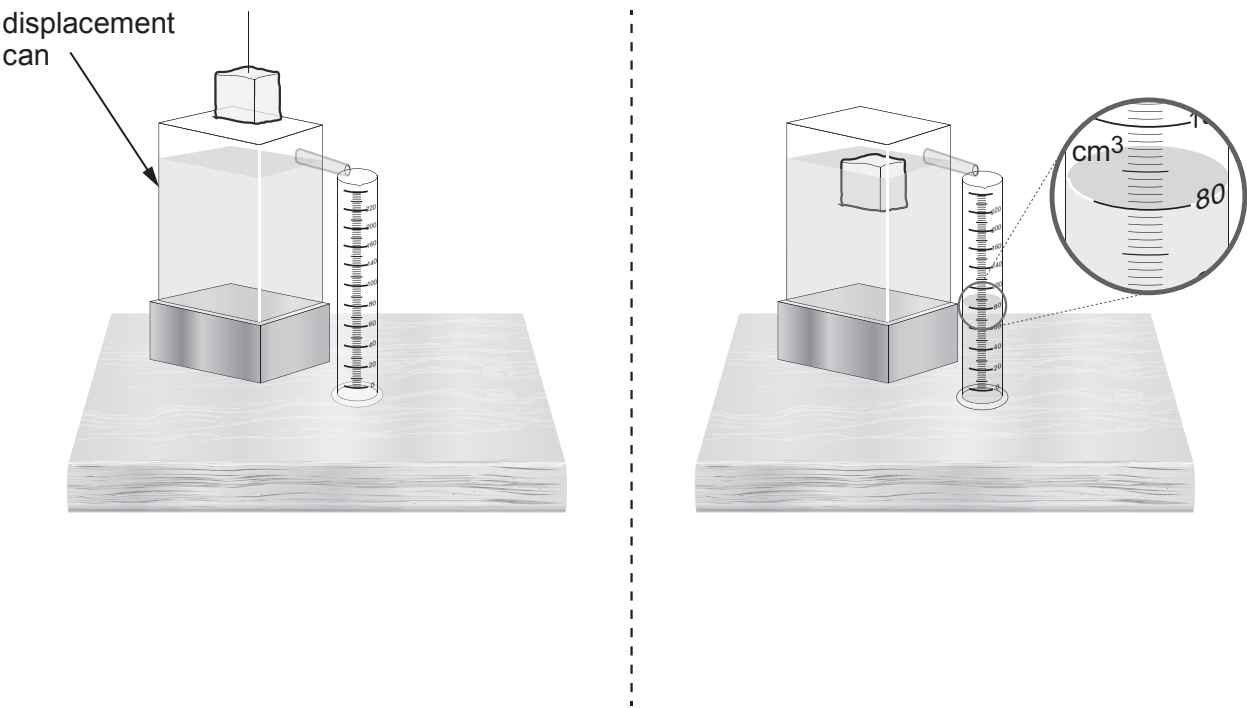
9. (a) (i) State Archimedes' principle. [1]

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(ii) A student lowers a block of ice into a displacement can containing salt water. As shown in the diagram, 80 cm³ of displaced salt water is collected in the measuring cylinder.



Calculate the mass of salt water displaced taking the density of salt water, $\rho_{\text{salt water}}$, to be 1030 kg m⁻³. [1 cm³ = 1 × 10⁻⁶ m³] [2]

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(iii) The density of ice, ρ_{ice} , is 920 kg m^{-3} . Use this and the Archimedes' principle to determine the volume of the block of ice above the surface. [3]

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(iv) The Greenland ice sheet is a vast body of ice covering roughly 80 % of the surface of Greenland. It is estimated that Greenland's melting ice is releasing $2.2 \times 10^{11} \text{ m}^3$ of water into the surrounding ocean every year.

I. If the surface area of oceans on Earth is $3.6 \times 10^{14} \text{ m}^2$ calculate the rise in water levels per year from Greenland ice melting. [1]

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II. Explain why the melting of the ice sheet would have a greater effect on global sea levels than the melting of the same mass of icebergs. [2]

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III. It is believed that melting ice sheets may lead to further increases in global temperatures. Suggest a reason for this. [1]

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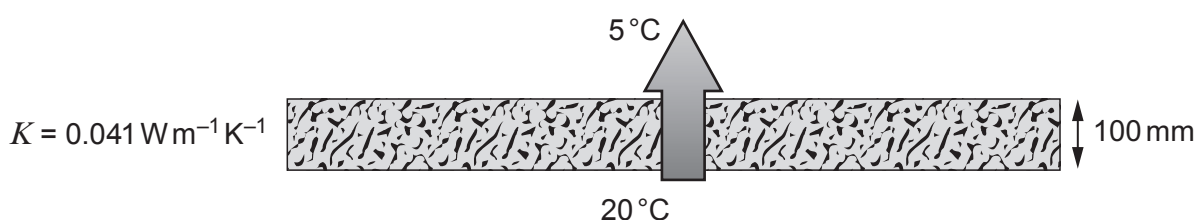
- (b) (i) Use an appropriate equation to show that the unit of the coefficient of thermal conductivity, K , is $\text{W m}^{-1} \text{K}^{-1}$. [2]

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- (ii) The recommended thickness of loft insulation has changed over time. In 1985, a 100 mm layer of fibre glass loft insulation ($K = 0.041 \text{ W m}^{-1} \text{K}^{-1}$) was used to cover an area of 72 m^2 in the roof space of a house. During winter, the air temperature just above the insulation was 5°C and the temperature of the surface supporting the insulation was 20°C . Calculate the rate of heat flow through the insulation. [2]

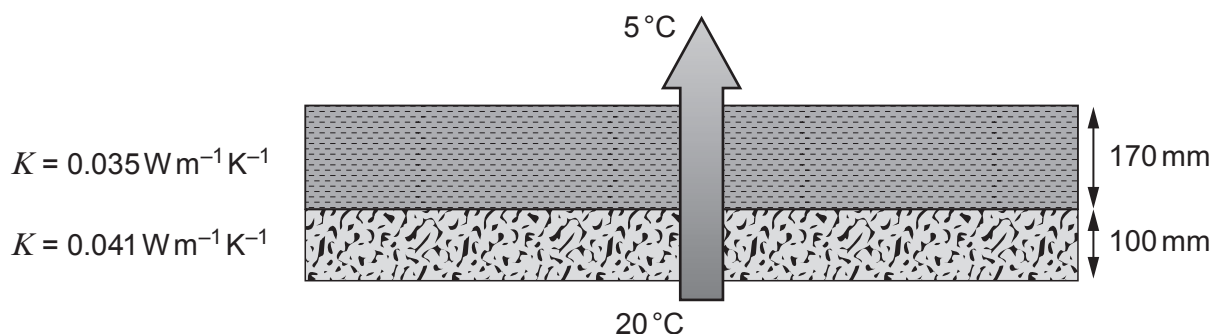


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- (iii) A present day loft insulation manufacturer recommends a loft insulation thickness of 270 mm. They suggest the house could achieve this modern standard by adding 170 mm of cellulose loft insulation ($K = 0.035 \text{ W m}^{-1} \text{K}^{-1}$) and that it would reduce the rate at which energy is transferred by more than 60%.



Again, taking the air temperature just above the insulation to be 5°C and the temperature of the surface supporting the insulation as 20°C , investigate whether their recommendation is correct. [4]

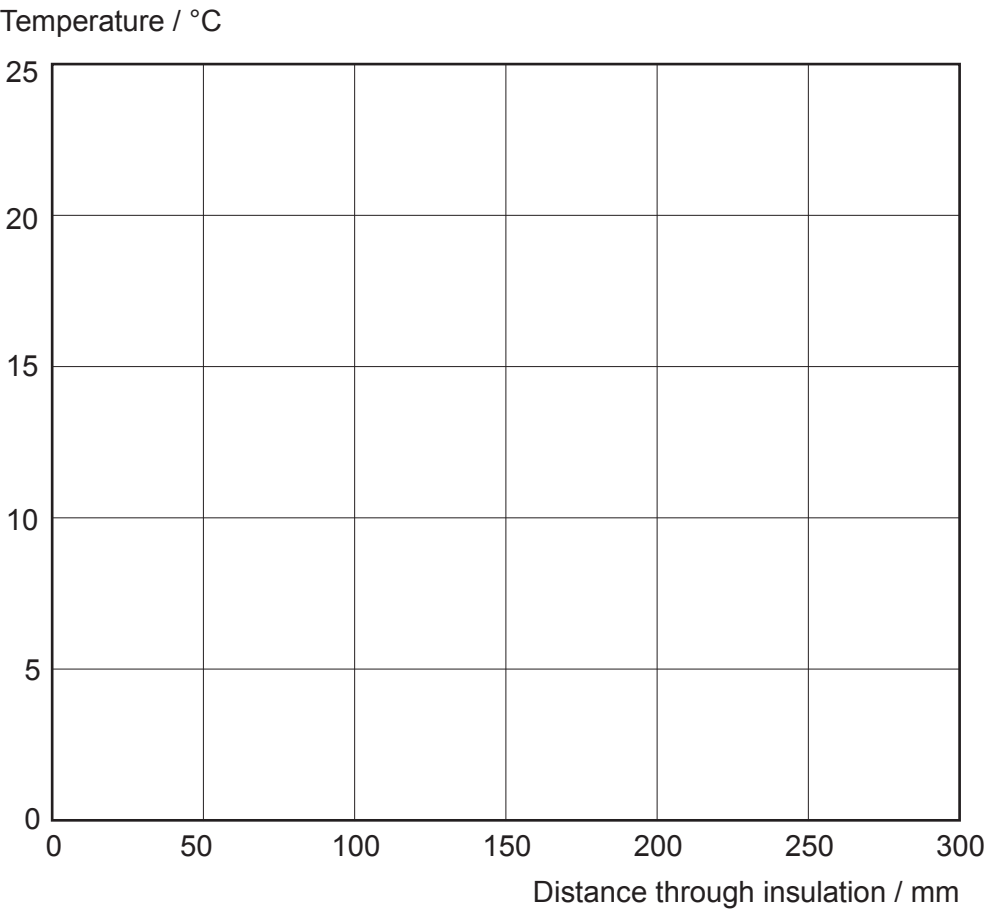
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- (iv) Starting from the lower surface at 20°C, sketch a graph of temperature against distance through the loft insulation on the axis provided. [2]



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